

Requirement Analysis – 4. Technology Stack (Architecture & Stack)

Incident Lifecycle Automation using ServiceNow

Architecture Overview

The solution is implemented within the ServiceNow platform. The architecture connects the user interface, Incident Management records, automation and workflow components, knowledge, change and related-record tracking.

Layer	Technology / ServiceNow Component	Purpose
User / Experience Layer	Service Operations Workspace	Create, view and manage incidents and related records.
ITSM Application Layer	Incident Management	Manage incident creation, classification, assignment, escalation, resolution and lifecycle states.
Automation Layer	Flow Designer / platform automation	Support automated workflow actions, notifications and related process activities.
Knowledge Layer	Knowledge Management / Agent Assist	Surface relevant knowledge articles and create reusable knowledge from resolved incidents.
Change Layer	Change Management	Create or track emergency changes related to incidents.
Data / Record Layer	ServiceNow tables and related records	Store incident, SLA, change, child incident and knowledge information.
Access & Control	ServiceNow roles and permissions	Control access to records and actions according to user responsibilities.

Technology Stack

Category	Selected Technology / Platform
Platform	ServiceNow
Application Area	Incident Management / ITSM
User Interface	Service Operations Workspace
Workflow / Automation	Flow Designer and ServiceNow platform automation
Knowledge	ServiceNow Knowledge Management with Agent Assist
Change Integration	ServiceNow Change Management
Data Storage	ServiceNow platform database / application tables
SLA Tracking	ServiceNow Task SLA / related records
Access Control	ServiceNow roles and permissions

Note: The stack above is based on the ServiceNow components used or specified in the project material; it avoids introducing unrelated external technologies.